8 The probability function of a discrete random variable X is

$$P(X = r) = \frac{k}{r}, r = 1,2,3,4,$$

where k is a constant.

(a) Determine the value of k. **[2]**

Two independent readings of X are taken. These are denoted by X_1 and X_2.

(b) Determine $P(X_1 + X_2 = 7)$. **[2]**